ONLINE BUS PASS APPLICATION SYSTEM

AI23431-WEB TECHNOLOGY AND MOBILE APPLICATION

MINI-PROJECT REPORT

SUBMITTED BY

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(241801100)

in partial fulfillment for the award of the degree of

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DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

RAJALAKSHMI ENGINEERING COLLEGE

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RAJALAKSHMI ENGINEERING COLLEGE

(an Autonomous Institution Affiliated to Anna University, Chennai)

BONAFIDE CERTIFICATE

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DEPARTMENT VISION

To promote highly Ethical and Innovative Computer Professionals through excellence in teaching, training and research.

DEPARTMENT MISSION

- To produce globally competent professionals who are motivated to learn emerging technologies and demonstrate innovation in solving real-world problems.
- To promote research activities among students and faculty members that contribute meaningfully to society.
- To instill strong moral and ethical values in professional practice.

PROGRAMME EDUCATIONAL OBJECTIVES(PEO'S)

PEO 1: To equip students with an essential background in computer science, basic electronics and applied mathematics.

PEO 2: To prepare students with fundamental knowledge in programming languages, and tools and enable them to develop applications.

PEO 3: To encourage the research abilities and innovative project development in the field of AI, ML, DL, networking, security, web development, Data Science and also emerging technologies for the cause of social benefit.

PEO 4: To develop professionally ethical individuals enhanced with analytical skills, communication skills and organizing ability to meet industry.

PROGRAMME OUTCOMES (POs)

PO 1: Engineering knowledge: Apply the knowledge of Mathematics, Science, Engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO 2: Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO 3: Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO 4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO 5: Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO 6: The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO 7: Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO 8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO 9: Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO 10: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO 11: Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO 12: Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOs)

A graduate of the Artificial Intelligence and Data Science Program will demonstrate

PSO 1: Foundation Skills: Ability to understand, analyze and develop computer programs in the areas related to algorithms, system software, web design, AI, machine learning, deep learning, data science, and networking

PSO 2: Problem-Solving Skills: Ability to apply mathematical methodologies to solve computational task, model real world problem using appropriate AI and ML algorithms. To understand the standard practices and strategies in project development, using open- ended programming environments to deliver a quality product.

PSO 3: Successful Progression: Ability to apply knowledge in various domains to identify research gaps and to provide solution to new ideas, inculcate passion towards higher studies, creating innovative career paths to be an entrepreneur and evolve as an ethically social responsible AI and ML professional.

COURSE OBJECTIVE

- To provide foundational knowledge and practical skills in creating and structuring web pages using HTML, enabling students to build accessible and well-organized websites.
- To understand and practice Embedded Dynamic Scripting on Client-side Internet Programming.
- To implement Server Side Scripting.
- To facilitate students to understand android Application Design
- To help students to gain a basic understanding of Android application development

COURSE OUTCOME

On completion of course students will be able to,

- CO 2: Upon completing the course, students will be able to design and develop basic web pages with proper structure and semantic elements, ensuring accessibility and functionality.
- CO 3: Design and implement dynamic web page with validation using javascript objects and by applying different event handling mechanism.
- CO 4: Design and implement simple webpage to learn JSP and Servlet.
- CO 5: Design and implement simple Application Design.
- CO 6: Identify various concepts of mobile programming that make it unique from programming for other

CO-PO-PSO Mapping

CO	PO 1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	3	3	2	2	2	3	2	3	3	3	3	3
CO2	3	3	3	3	3	2	-	-	2	2	2	3	3	2	2
CO3	3	3	3	2	3	1	1	2	3	3	3	3	3	3	3
CO4	3	3	3	3	3	2	1	2	3	2	2	3	3	3	3
CO5	1	1	1	1	1	-	-	-	3	3	3	3	1	-	2

Note: Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3:

Substantial (High) No correlation: "-"

SUSTAINABLE DEVELOPMENT GOALS

SDG	GOAL	JUSTIFICATION
SDG-4	Quality Education	Education drives progress on other SDGs, breaks poverty cycles, and promotes peace and inequality reduction.
SDG-8	Decent Work and Economic Growth	Necessary for social stability, shared prosperity, and combating poverty.
SDG-9	Industry, Innovation, and Infrastructure	Infrastructure is vital for economic growth and stability. It acts as a poverty reduction tool through job creation and provides technological solutions to environmental challenges.
SDG-17	Partnerships for the Goals	Given the interconnected nature of the SDGs, success requires collaborative action, global resource sharing, and bridging the immense financing gap faced by developing nations.

ABSTRACT

The Online Bus Pass Application System is a web-based application developed to simplify the process of applying, renewing, and managing bus passes. The system helps students, employees, and passengers apply for bus passes online without visiting transport offices directly. It reduces manual paperwork, saves time, and improves efficiency in managing bus pass records.

In this system, users can register using their personal details, log in securely, fill out the application form, upload required documents, and submit the request online. The administrator verifies the details and approves or rejects the application. After approval, users can download or print their digital bus pass.

The system maintains a database for storing user information, payment details, application status, and pass validity. It also provides features like pass renewal, status tracking, and notifications. The application is developed using technologies such as HTML, CSS, JavaScript, PHP, and MySQL.

The main objective of this project is to provide a fast, secure, accurate, and user-friendly platform for bus pass management while reducing workload for transport authorities and improving convenience for passengers..

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
HTML	HyperText Markup Language
CSS	Cascading Style Sheets
JSP	Java Server Pages
JDBC	Java Database Connectivity
SQL	Structured Query Language
DBMS	Database Management System
UI	User Interface
UX	User Experience
HTTP	HyperText Transfer Protocol

CHAPTER 1

INTRODUCTION

1.1 General

The Online Bus Pass Application System is a web-based application developed to simplify the process of applying, renewing, and managing bus passes digitally.

It helps students, employees, and regular passengers apply for bus passes without visiting transport offices manually.

The system allows users to register, log in, fill in application details, upload required documents, and make online payments.

After verification, the bus pass is generated digitally and can be downloaded or printed by the user. Administrators can manage user details, verify applications, approve or reject requests, and maintain records efficiently

1.2 Need for online bus pass application system

Traditional methods of bus pass application system retrieval are often unreliable, time-delayed, and limited in coverage. People who travel, farm, or plan outdoor activities need accurate and up-to-date weather data. General broadcasts may not cover specific locations, leaving users with insufficient information.

An online bus pass application system addresses these challenges by providing a centralized platform where multiple users can interact. It reduces dependency on brokers, improves accessibility, and allows users to compare different weather locations easily. Additionally, it ensures that information is updated in real time, making the process more reliable and efficient.

1.3 Scope of the System

The system is designed to serve as a comprehensive platform for weather data retrieval and display. It allows users to register, log in, and manage their personalized weather dashboards. Users can search for weather by city name or coordinates, view current conditions including temperature, humidity, wind speed, and UV index. The system also provides a 7-day forecast to help users plan ahead.

Furthermore, users can save favourite locations and receive weather alerts for severe conditions such as storms or extreme temperatures

1.4 Organization of the Report

This report is organized into several chapters that describe the development and implementation of the online bus pass application system.

Chapter 1: provides the introduction to the project, including the general overview, need for the system, and scope of the project.

Chapter 2: presents the literature survey, which discusses existing weather data platforms and related work.

Chapter 3: explains the problem formulation and objectives of the proposed system.

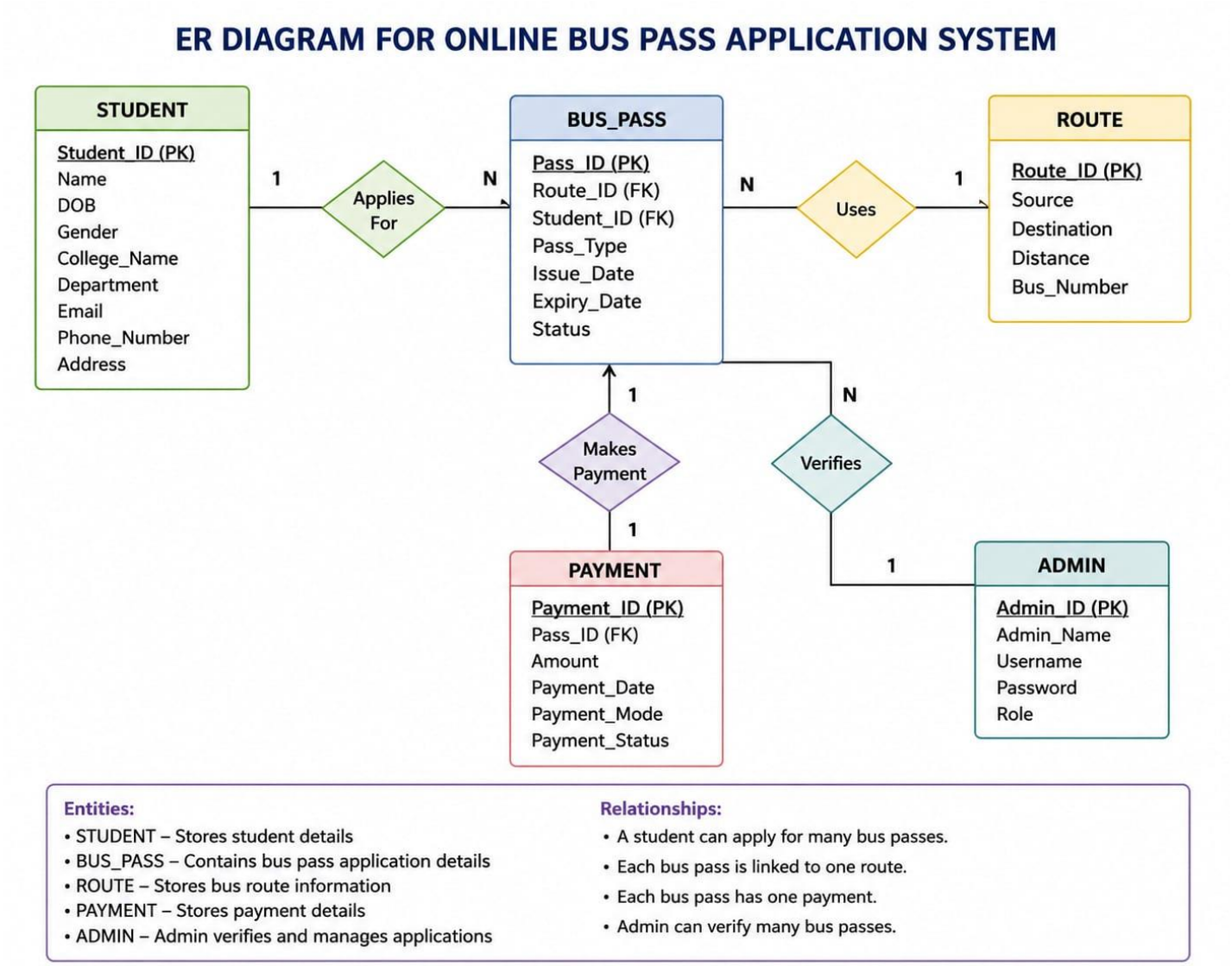
Chapter 4: describes the system design, including system architecture, database design, and system requirements.

Chapter 5: discusses the implementation of the system and explains the different modules.

Chapter 6: presents the results and discussion, including system outputs and performance evaluation.

Chapter 7: concludes the project and suggests possible future enhancements.

System architecture diagram:s



Entities:

- STUDENT – Stores student details
- BUS_PASS – Contains bus pass application details
- ROUTE – Stores bus route information
- PAYMENT – Stores payment details
- ADMIN – Admin verifies and manages applications

Relationships:

- A student can apply for many bus passes.
- Each bus pass is linked to one route.
- Each bus pass has one payment.
- Admin can verify many bus passes.

CHAPTER 2

LITERATURE SURVEY

2.1 Introduction

The literature survey explores existing systems and approaches used in weather information retrieval and display platforms. With the rapid growth of internet usage, digital solutions have become the preferred method for accessing weather data. This section examines how current platforms function, the technologies they use, and the challenges they face. Understanding these aspects helps in designing a system that is both efficient and user-friendly while addressing the limitations of existing solutions.

2.2 Existing Online Bus Pass Application Systems

The existing online bus pass application system is a digital platform used by students, employees, and the public to apply for bus passes through the internet instead of visiting transport offices manually.

The system helps users register, fill in application forms, upload documents, and renew passes online.

In the current system, users must first create an account and log in to the portal. After logging in, they can select the type of bus pass, enter personal details, upload required documents such as ID proof or student certificate, and submit the application.

The transport authority verifies the details and approves or rejects the request. Once approved, the user can download or collect the bus pass.

2.3 Related Work

Previous studies and projects in the domain of web-based weather information systems have focused on improving usability, efficiency, and data accuracy. Researchers have explored the

systems. Database-driven applications are commonly used to cache and manage large volumes of weather data, ensuring that users can access updated and accurate information.

Security and user authentication are also key areas of focus in related work, with systems implementing login mechanisms and session management to protect user data. Some solutions incorporate advanced features such as weather prediction models and real-time radar integration to enhance user experience. However, these features often increase system complexity and may not be necessary for all users, especially in smaller-scale applications.

2.4 Summary

The analysis of existing platforms and related work highlights the importance of providing a balanced system that combines functionality with simplicity. While many advanced systems offer a wide range of meteorological features, there is still a need for applications that are easy to use, efficient, and focused on essential weather functionalities.

Based on this study, the proposed system aims to deliver a streamlined weather information platform that enables users to browse, search, and manage weather data with ease. The focus is on creating a reliable and user-friendly application that meets core requirements without unnecessary complexity

CHAPTER 3

PROBLEM FORMULATION AND OBJECTIVES

3.1 Problem Statement

In the current digital environment, accessing accurate and real-time online bus pass application system is often challenging. Users may struggle to find location-specific forecasts, while agriculture, tourism, and transport sectors face significant challenges due to lack of timely weather updates.

Existing weather platforms, although feature-rich, are often complex and may include unnecessary functionalities for basic users. They can also rely heavily on expensive commercial APIs, which may increase operational costs and reduce data availability for smaller applications. Additionally, many systems do not provide a simple and centralized interface for managing saved locations and personalized weather preferences.

Therefore, there is a need for a streamlined and user-friendly web-based system that allows users to easily search, view, and manage weather information for multiple locations, while providing timely and accurate meteorological data in an accessible format.

3.2 Objectives of the System

The main objective of the system is to develop a web-based application that simplifies the process of accessing and managing weather information. The specific objectives are as follows:

- To design and develop a user-friendly interface for viewing weather information and forecasts
- To enable users to search weather data based on city name or geographic coordinates
- To provide functionality for users to save and manage their favourite locations
- To allow users to view detailed meteorological data including temperature, humidity, wind speed, and UV index
- To implement a weather alert feature to notify users of severe weather conditions
- To develop a secure login and registration system using session management
- To create a structured and maintainable system using appropriate design patterns and API integration techniques

CHAPTER 4

SYSTEM DESIGN

4.1 Introduction

System design is an important phase in software development where the overall structure and components of the system are planned. It defines how different modules interact with each other and how data flows within the system. A well-designed system improves efficiency, reliability, and ease of maintenance.

The weather information system is designed as a web-based application that allows users to browse, search, and manage bus pass application. It provides features such as user registration, login, property posting, and inquiry handling. The system follows a client-server architecture where users interact through web pages, while the backend processes requests and communicates with the database.

4.2 System Architecture

The system follows a three-layer architecture consisting of the presentation layer, application layer, and data layer.

Presentation Layer

This layer represents the user interface where users interact with the system. It includes web pages such as the homepage, login page, registration page, weather dashboard page, and location search page. These pages are developed using JSP, HTML, and CSS, and are responsible for displaying weather data to users and collecting search input.

Application Layer

This layer handles the core logic of the system. It processes user requests, manages sessions, fetches data from the weather API, performs validation, and controls the flow of data between the presentation layer and the database.

The backend is implemented using Java Servlets, where each servlet is responsible for a specific functionality:

- PropertyServlet → Displays all weather informations
- SearchServlet → Handles property search
- AddPropertyServlet → Adds new weather informations
- PropertyDetailServlet → Shows detailed property information
- InquiryServlet → Manages user inquiries
- LoginServlet & RegisterServlet → Handle authentication
- DeletePropertyServlet → Deletes weather informations

This layer ensures proper execution of all operations and maintains system functionality.

Data Layer

This layer is responsible for storing and managing all application data. The system uses a MySQL database connected through JDBC for data operations, complemented by a weather API for real-time meteorological data.
The database consists of three main tables:

- **users** → Stores user details such as name, email, and password
- **weather locations** → Stores property information including title, price, location, description, image, and user ID
- **inquiries** → Stores user inquiries related to weather locations

4.3 Database Design

The database is designed to efficiently store and manage data related to users, saved locations, and cached bus information. Each table is structured with appropriate fields and relationships.

- The **users** table stores registered user information
- The **bus locations** table stores all bus informations and is linked to users through a foreign key
- The **inquiries** table stores messages sent by users regarding specific bus locations

4.4 System Requirements

System requirements define the hardware and software resources needed to run the application successfully.

4.4.1 Hardware Requirements

The system does not require high-end hardware and can run on standard computers.

- **Processor:** Intel Core i3 or higher
- **RAM:** Minimum 4 GB
- **Storage:** 500 GB Hard Disk or SSD
- **Input Devices:** Keyboard and Mouse
- **Internet Connection:** Required

4.2.1 Software Requirements

The system is developed using commonly available tools and technologies.

5 **Operating System:** Windows 10 or later

6 **Frontend Technologies:** HTML, CSS, JSP

7 **Backend Technology:** Java Servlets

8 **Database:** MySQL

9 **Database Connectivity:** JDBC

10 **Web Server:** Apache Tomcat 9.0

11 **Development Environment:** Eclipse IDE for Enterprise Java

12 **Web Browser:** Google Chrome or any modern browser

CHAPTER 5

SYSTEM IMPLEMENTATION

5.1 System Methodology

The system is developed using a structured approach based on the Model-View-Controller (MVC) architecture. This methodology helps in separating the user interface, business logic, and data handling, making the system easier to develop and maintain.

- **Model:** Represents the data and business logic using Java classes such as Property, User, and Inquiry.
- **View:** Consists of JSP pages that display data to users and collect input.
- **Controller:** Implemented using Java Servlets, which handle user requests, process data, and control the flow between the view and model.

The development process involves designing the database, integrating the weather API, creating model classes, implementing servlets for different functionalities, and developing JSP pages for user interaction. JDBC is used to connect the application with the MySQL database.

5.2 Module Descriptions

The system is divided into several modules, each responsible for a specific functionality.

5.2.1 User Module

This module handles user registration, login, and session management.

- New users can register by providing details such as name, email, and password
- Existing users can log in using valid credentials
- Session management is implemented using HttpSession
- User data is stored securely in the database

5.2.2 Location Management Module

This module allows users to save, view, and delete favourite weather locations.

- Users can save new locations by entering city name or allowing GPS-based detection
- Saved locations are stored in the database for quick access
- Users can delete their saved locations
- All saved locations with current weather are displayed on the dashboard

5.2.3 Weather Search Module

This module provides functionality to search for weather data based on city name.

- Users can enter a city name or postal code
- The system performs a search using SQL LIKE queries
- Current weather conditions and forecasts are displayed dynamically
- Helps users quickly access weather information for any location

5.2.4 Weather Details Module

This module displays detailed meteorological information for a selected location.

- Shows complete details such as temperature, humidity, wind speed, pressure, UV index, and visibility
- Allows users to view full weather conditions and 7-day forecast before planning activities
- Improves user experience by providing detailed and accurate weather data

5.2.5 Weather Alert Module

This module enables users to set and manage weather alerts.

- Users can configure alerts for specific weather conditions such as rain, storm, or extreme temperature
- Alert preferences are stored in the database
- Helps users stay informed about severe or changing weather conditions

5.2.6 Login and Authentication Module

This module ensures secure access to the system.

- Validates user credentials during login
- Uses session management to maintain user state
- Prevents unauthorized access to restricted features
- Handles login and logout operations

5.2.7 Delete Location Module

This module allows users to remove their saved weather locations.

- Only the account owner can delete their saved locations
- Related cached weather data is also cleaned up from the database
- Maintains data consistency and integrity

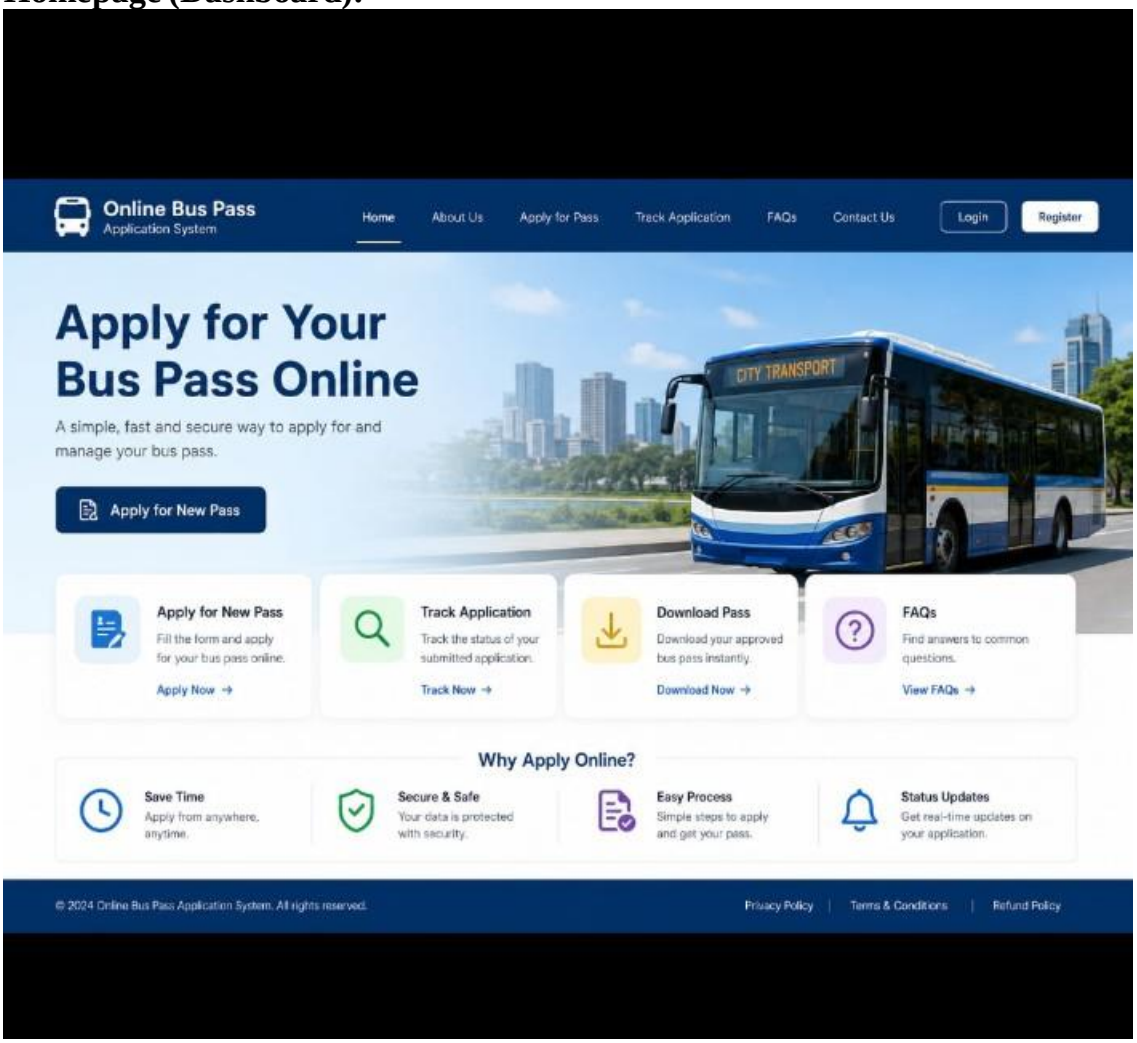
CHAPTER 6

RESULTS AND DISCUSSION

6.1 System Output Screens

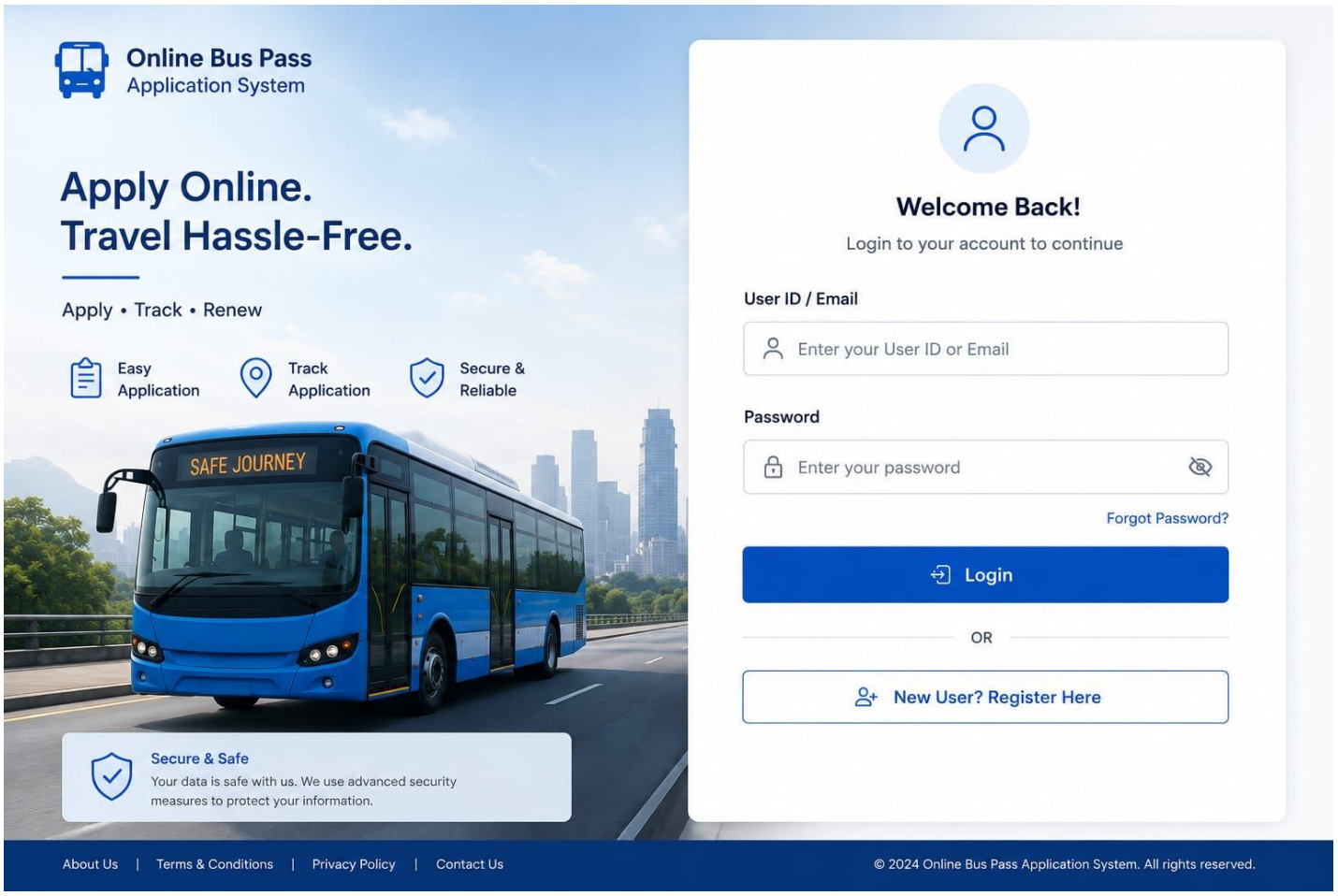
The developed system was tested by executing all major functionalities, and the outputs were verified through different user interactions. The application successfully displays real-time weather data fetched from the API and responds correctly to user inputs.

The main output screens of the system include:

- **Homepage (Dashboard):**
- 

- **Login Page:**

  Allows registered users to log in using their email and password. Proper validation messages are displayed for incorrect credentials.



Online Bus Pass
Application System

Home | Help & Support | Login

Apply Online. Travel Hassle-Free.

Easy Application | Secure | Reliable

Easy Application
Simple and quick online application process.

Secure & Safe
Your data is safe with us. We protect your information.

Track Application
Track your application status online anytime.

Create New Account

Register to apply for your bus pass online

Personal Information

Full Name *
Enter your full name

Date of Birth *
DD/MM/YYYY

Gender *
Select Gender

Mobile Number *
Enter 10 digit mobile number

Email ID *
Enter your email address

Address Information

Address Line 1 *
House No., Street, Area

Address Line 2
Landmark (Optional)

City / Town *
Enter your city

State *
Select State

Pincode *
Enter pincode

Account Information

User ID *
Choose a User ID

Password *
Create a password

Confirm Password *
Confirm your password

☐ I agree to the Terms & Conditions and Privacy Policy.

← Back to Login | Register Now

Note: All fields marked with * are mandatory.

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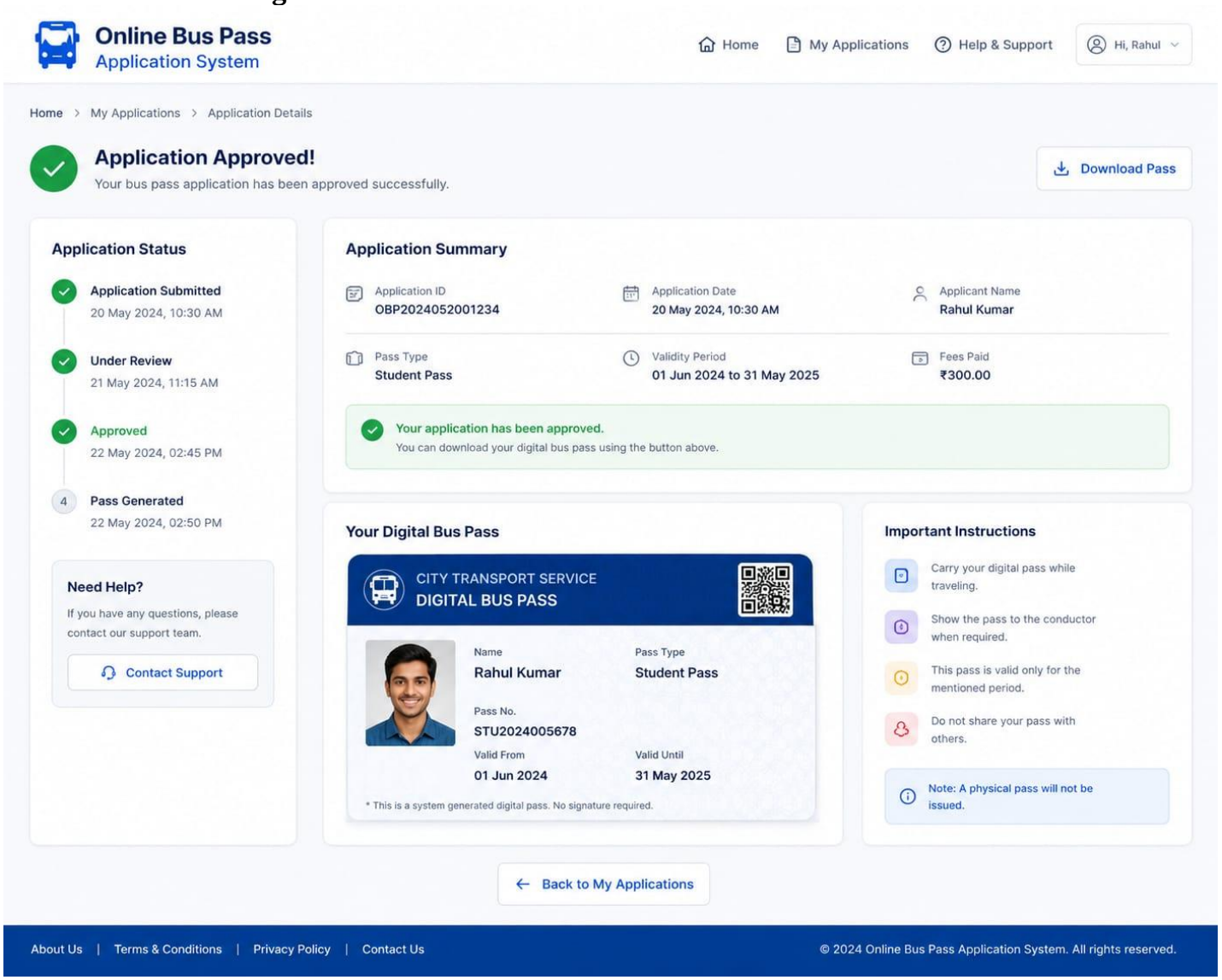
- **Results Page:**



- **Details Page:**

- Dashboard
- My Applications
- Profile
- Change Password
- Help & Support
- Logout

Home My Applications > Application Details

Application Details

View your bus pass application information and status

Download Pass | Print

Application Status

- **Application Submitted** — 20 May 2024, 10:30 AM
- **Under Review** — 21 May 2024, 11:15 AM
- **Approved** — 22 May 2024, 02:45 PM
- 4 **Pass Generated** — 22 May 2024, 02:50 PM

Application Summary

Application ID OBP2024052001234	Application Date 20 May 2024, 10:30 AM	Current Status Approved
Pass Type Student Pass	Validity Period 01 Jun 2024 to 31 May 2025	Fees Paid ₹300.00
Payment Method UPI	Transaction ID UPI521456987123	Payment Date 20 May 2024, 10:28 AM

Quick Actions

- Download Pass
- Print Pass
- Share Pass
- Cancel Application

Your Digital Bus Pass

CITY TRANSPORT SERVICE
DIGITAL BUS PASS

Name: Rahul Kumar
Pass Type: Student Pass
Pass No.: STU2024005678
Valid From: 01 Jun 2024
Valid Until: 31 May 2025

* This is a system generated digital pass. No signature required.

Personal Information

Full Name	Rahul Kumar
Date of Birth	15 Aug 2004
Gender	Male
Mobile Number	9876543210
Email ID	rahul.kumar@email.com
Address	H.No. 123, Green Park, Sector 12, Lucknow, Uttar Pradesh - 226012
City / Town	Lucknow
State	Uttar Pradesh
Pincode	226012

Important Instructions

- Carry your digital pass while traveling.
- Show the pass to the conductor when required.
- This pass is valid only for the mentioned period.
- Do not share your pass with others.

← Back to My Applications

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These output screens demonstrate that the system is functioning correctly and providing the intended features.

6.2 Performance Evaluation

The system was evaluated based on responsiveness, accuracy, and usability.

- **Response Time:**
  The system responds quickly to user requests such as loading weather data, searching cities, and displaying detailed forecasts.
- **Data Accuracy:**
  All weather data retrieved from the API and database is displayed correctly without inconsistencies.
- **Usability:**
  The user interface is simple and easy to navigate, allowing users to access weather information without confusion.
- **Efficiency:**
  The use of JDBC, optimized SQL queries, and efficient API calls ensures efficient data handling and retrieval.

Overall, the system performs well for small to medium-scale usage and provides a smooth user experience for accessing weather data.

6.3 System Testing

The system was tested using different scenarios to ensure reliability and correctness.

- **Functional Testing:**
  All features such as registration, login, location saving, weather search, forecast display, and alerts were tested and found to be working as expected.
- **Validation Testing:**
  Input fields were tested for correct validation, ensuring that invalid or empty data is not accepted.
- **Authentication Testing:**

Login and session management were tested to ensure secure access to the system.

- **Database Testing:**

  Data insertion, retrieval, and deletion operations were verified to ensure proper interaction with the database.

- **Error Handling:**

  The system handles errors gracefully and displays appropriate messages to the user.

The testing results confirm that the system is stable, reliable, and ready for deployment.

CHAPTER 7

CONCLUSION AND FUTURE WORK

7.1 Conclusion

The online bus pass application system was successfully designed and developed to simplify the process of searching, adding, and managing weather informations. The application provides a user-friendly interface that allows users to browse weather locations, search based on location, view detailed information, and communicate with weather data sources through inquiries.

The system effectively integrates frontend and backend technologies using Java Servlets, JSP, MySQL, and the OpenWeatherMap API. It follows a structured architecture that ensures clear separation of components, making the application easy to understand and maintain. Key features such as user authentication, location management, weather API integration, and database interaction have been implemented successfully.

The project demonstrates the practical application of web development concepts and provides a reliable platform for accessing real-time weather information. Overall, the system meets its objectives and performs efficiently for small to medium-scale usage.

7.2 Future Enhancements

1. Mobile Application Integration

Develop Android and iOS apps for easier access and faster applications.

2. Online Payment Gateway

Enable secure online payments using UPI, debit/credit cards, and net banking.

3. QR Code / Smart Card Bus Pass

Generate digital passes with QR codes for quick verification by bus conductors.

4. GPS Bus Tracking

Allow students/passengers to track bus locations in real time.

5. SMS and Email Notifications

Send alerts for application approval, renewal dates, and payment confirmation.

6. Automatic Pass Renewal

Provide auto-renewal options for eligible users.

7. AI-based Verification System

Use AI to detect fake documents and reduce manual verification work.

8. Multi-language Support

Add regional language support for better user experience.

9. Cloud Database Storage

Improve scalability, security, and backup using cloud technology.

10. Analytics and Report Generation

Generate detailed reports for administrators to monitor pass usage and revenue.

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APPENDICES

Appendix-1: Sample Database Tables

Users Table

Field Name	Data Type	Description
user_id	INT (PK)	Unique user ID
name	VARCHAR	User name
email	VARCHAR	User email
password	VARCHAR	User password

Saved Locations Table

Field Name	Data Type	Description
location_id	INT (PK)	Unique location ID
title	VARCHAR	City name
price	DOUBLE	Latitude coordinate
location	VARCHAR	Longitude coordinate
description	TEXT	Country code
image	VARCHAR	Timezone offset
user_id	INT (FK)	Reference to user

Weather Cache Table

Field Name	Data Type	Description
cache_id	INT (PK)	Unique cache ID
location_id	INT (FK)	Related location
name	VARCHAR	Weather condition
message	TEXT	JSON data from API

Appendix-2: Sample SQL Queries

```
-- Insert User
INSERT INTO users (name, email, password)
VALUES ('John Doe', 'john@example.com', '123456');

-- Insert Saved Location
INSERT INTO weather locations (title, price, location, description, image, user_id)
VALUES ('2BHK Apartment', 5000000, 'Chennai', 'Spacious flat', 'img1.jpg', 1);

-- Search Saved Location
SELECT * FROM weather locations
WHERE location LIKE
'%Chennai%';

-- Delete Location
DELETE FROM
saved_locations WHERE
location_id = 1;
```

Appendix-4: Key Servlet Example

```
@WebServlet("/index")
public class WeatherServlet extends HttpServlet {

    protected void doGet(HttpServletRequest request,
                         HttpServletResponse response)
            throws ServletException, IOException {

        List<WeatherData> list = new ArrayList<>();
        Connection con = DBConnection.getConnection();

        try {
            Statement stmt = con.createStatement();
            ResultSet rs = stmt.executeQuery(
                "SELECT * FROM weather locations");

            while (rs.next()) {
                Property p = new Property();
                p.setTitle(rs.getString("title"));
                p.setPrice(rs.getDouble("price"));
                list.add(p);
            }

        } catch (Exception e) {
            e.printStackTrace();
        }
```

```
        request.setAttribute("propertyList", list);
        request.getRequestDispatcher("index.jsp")
                .forward(request, response);
    }
}
```

PSEUDOCODE

The following pseudocode represents the core functionalities of the weather information system.

1. User Registration

```
BEGIN
  INPUT name, email, password

  CONNECT to database

  CHECK if email already exists
  IF exists THEN
     DISPLAY "Email already registered"
  ELSE
     INSERT user details into users table
     DISPLAY "Registration successful"
  ENDIF

  CLOSE database connection
END
```

2. User Login

```
BEGIN
  INPUT email, password

  CONNECT to database

  FETCH user where email and password match

  IF user exists THEN
     CREATE session
     STORE user details in session
     REDIRECT to homepage
  ELSE
     DISPLAY "Invalid
  credentials" ENDIF

  CLOSE database connection
END
```

3. Save Location

```
BEGIN
  CHECK if user is logged in

  INPUT city_name, latitude, longitude CONNECT to

  database

  INSERT location details into saved_locations
      table with user_id from session
  DISPLAY "Property added successfully"
  CLOSE database connection
END
```

4. View Weather Dashboard

```
BEGIN
  CONNECT to database

  FETCH all saved locations FETCH current

  weather for each location from API STORE

  results in weatherData object

  SEND list to JSP page

  DISPLAY weather dashboard to

  user

  CLOSE database connection
END
```

5. Search Weather by City

```
BEGIN
  INPUT city_name keyword

  CALL weather API with

  city_name

  PARSE API response:
      SELECT * FROM weather locations
      WHERE location LIKE '%keyword%'

  STORE results in weatherData object
```

```
  DISPLAY current weather and forecast to user

  CLOSE database connection
END
```

6. View Weather Details

```
BEGIN
  GET city_name from request

  CALL weather API for city

  FETCH 7-day forecast from API DISPLAY

  complete weather details and forecast

  CLOSE database connection
END
```

7. Set Weather Alert

```
BEGIN
  INPUT alert_type, threshold, user_id

  CONNECT to database

  INSERT alert preferences into weather_alerts table
  DISPLAY "Inquiry sent successfully"
  CLOSE database connection
END
```

8. Delete Saved Location

```
BEGIN
  GET location_id

  CHECK if user is owner

  CONNECT to database

  DELETE related inquiries from inquiries table

  DELETE property from weather locations table

  DISPLAY "Property deleted successfully"
```

```
  CLOSE database connection
END
```

CERTIFICATE
of completion

cursa

We certify that the student:

JAYASREE D

completed the following course on our e-learning platform:

HTML

With a workload of 1 hour 55 minutes

Completed on April 29, 2026

Adrian Dantas - Executive Director - Cursa

jayasree

Student signature

This certificate proves the completion of the courses described herein on the Cursa platform, authenticated by the viewing record of the video lessons. You can verify its authenticity using the QR code or at cursa.app/pt/my-certificate, using the code on this certificate.

VALIDATION CODE:
707a48ff84368952f02ff174dcd87a27

Point the camera to verify authenticity

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CERTIFICATE
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We certify that the student:

JAYASREE D

completed the following course on our e-learning platform:

JavaScript for beginners

With a workload of 01 hours 05 minutes

Completed on April 29, 2026

Adrian Dantas - Executive Director - Cursa

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